



Thermostat Wiring: AC with Gas Furnace

Conventional system field reference | Brackett Heating, Air & Plumbing

SCOPE

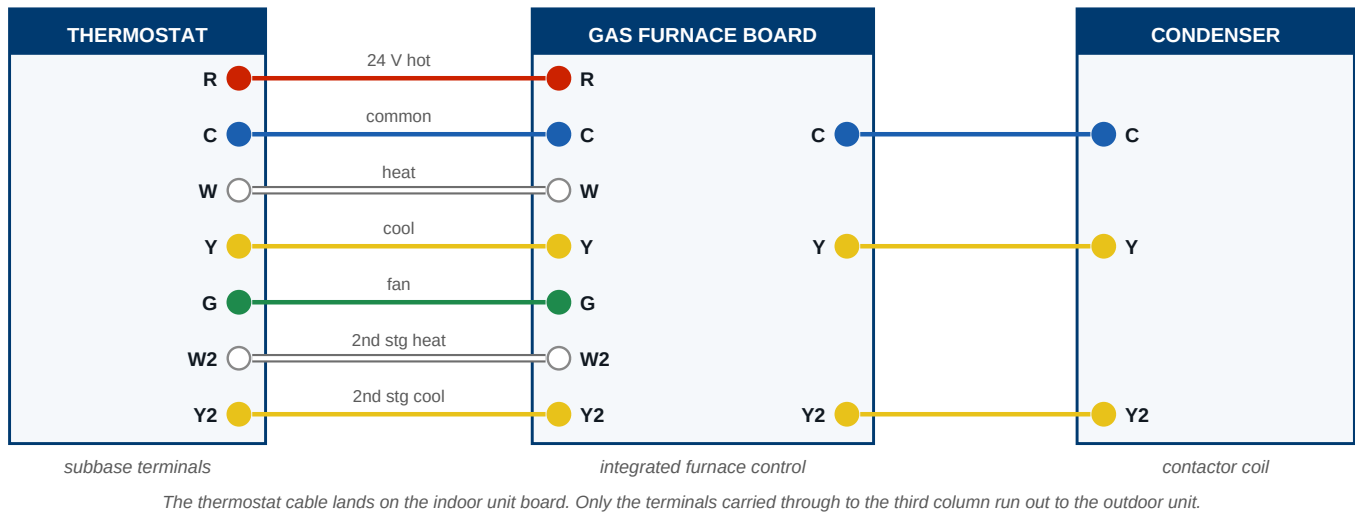
Single-stage or two-stage air conditioning paired with a gas furnace. The furnace control board is the hub: the thermostat lands on the furnace, and the furnace feeds the condenser.

TERMINAL REFERENCE · AC + GAS FURNACE

Term	Common color	What it does	Lands on
R	Red	24 VAC hot from the furnace control transformer	Furnace R. If the stat has Rc and Rh, jumper them together on a single-transformer system.
C	Blue (sometimes brown)	24 VAC common. Powers the thermostat.	Furnace C. Also runs to the condenser contactor.
W	White	Heat call. Energizes the furnace heat sequence.	Furnace W / W1.
Y	Yellow	Cool call. Energizes the condenser contactor.	Furnace Y, which passes through to the condenser.
G	Green	Indoor blower.	Furnace G.
W2	Brown or pink	Second stage heat on a two-stage furnace.	Furnace W2. Omit on single stage.
Y2	Blue or violet	Second stage cooling on a two-stage condenser.	Furnace Y2 or direct to the condenser.

Rc / Rh: two R terminals exist for systems with separate heating and cooling transformers. On a furnace-plus-split-AC system there is one transformer, so the Rc-Rh jumper stays in. Removing it is a common cause of "cooling works, heat does nothing."

WIRING DIAGRAM · AC + GAS FURNACE



THE SHORT VERSION

R, C, W, G stop at the furnace. Y and C carry on out to the condenser contactor. W2 and Y2 only exist if the equipment is two-stage.

Thermostat configuration

- System type: conventional / gas-electric. Not heat pump.
- Heat stages: 1 or 2 to match the furnace. Cool stages: 1 or 2 to match the condenser.
- Fan with heat: **equipment controlled**. The furnace board runs its own blower timing on a heat call. Setting the stat to control the fan on heat will start the blower before the heat exchanger is warm and blow cold air.
- Fan with cool: thermostat controlled is normal.
- Cycles per hour / heat anticipator: leave at the default unless there is a specific complaint.

Safety and shop rules

- Kill power at the disconnect and the furnace service switch before landing or removing any wire. The 24 V side is still live if the transformer is powered.
- Shorting R to C blows the control transformer or the board fuse. Most boards carry a 3 A or 5 A automotive blade fuse - carry spares.
- Wire colors are convention, not code. Ring out every conductor before you trust it. On a change-out, label the old wires at both ends before you pull the old stat.
- Land a C wire on every modern thermostat. Power-stealing without a common causes intermittent faults that look like equipment problems.

Verification

- 1 Power up and confirm 24 to 28 VAC between R and C at the thermostat subbase.
- 2 Set the thermostat configuration to match the equipment before the first call (system type, stages, fan with heat).
- 3 Jumper test from the subbase with the thermostat removed: R to G should run the indoor blower only.
- 4 R to Y plus G should bring on cooling with the blower.
- 5 R to W should bring on heat.
- 6 Reinstall the thermostat and run a full cycle in every mode the system supports.
- 7 Verify the changeover actually changes over - feel the discharge air, do not trust the display.
- 8 Record the configuration settings on the invoice and inside the furnace door.

Common mistakes

Symptom	Check this first
Nothing works, blank stat	No C wire landed, or blown board fuse from an R-to-C short. Check the 3 A or 5 A blade fuse on the furnace board.
Cooling works, heat does nothing	Rc-Rh jumper removed on a single-transformer system.
Blower runs but no cooling outside	Y not carried from the furnace board out to the condenser, or the condenser disconnect is pulled.
Cold air at the start of a heat call	Thermostat set to control the fan on heat instead of the equipment.
Second stage never comes on	W2 not landed, or the stat is configured for one stage of heat.
Transformer keeps failing	A pinched thermostat cable, or a wire shorting in the condenser whip or at the contactor.